

Wearable system to help visually impaired navigate

The white cane that visually impaired people frequently use to navigate paths has two major drawbacks: One, the obstacles they come in contact with are sometimes other people. Two,



Navigation for visually impaired people (Image courtesy: <http://news.mit.edu>)

they can't identify certain types of objects, such as tables and chairs, or determine whether a chair is already occupied.

MIT Researchers have developed a new system that guides the user in the same way as a suite of sensors can guide a self-driving car. The system consists of a 3D camera worn in a pouch hung around the neck; a processing unit that runs the team's proprietary algorithms; a sensor belt, which has five vibrating motors evenly spaced around its forward half; and a reconfigurable Braille interface, which is worn at the user's side. The system could be used in conjunction with or as an alternative to a cane.

The key to the system is an algorithm for quickly identifying surfaces and their orientations from the 3D camera data. The algorithm first groups the pixels into clusters of three. Because the pixels have associated location data, each cluster determines a plane. If the orientations of the planes defined by five nearby clusters are within ten degrees of each other, the system concludes that it has found a surface and begins to buzz the associated motor if the wearer gets within two metres of it.

Telephone calls clear as a bell

Smartphones can do almost everything you want, but their poor voice quality is still a vexing issue. Fraunhofer researchers have helped develop a new codec that makes voice quality as natural as if the person you're calling is standing right next to you. That's because, for the first time, the entire audible frequency spectrum is transmitted.

The new Enhanced Voice Services (EVS) standard promises a step change comparable with the transition from analogue CRT to digital flat-screen TVs. Specifications for standards of this type are extremely demanding. "First of all, the codec must be capable of transmitting high-quality speech signals at relatively low data rates—so as not to compromise cost-efficiency," says Dipl.-Ing Markus Multus, who coordinated the software development part of the project at Fraunhofer IIS. Another requirement is that the codec should be sufficiently robust to recover from transmission errors, thereby ensuring that calls are not dropped due to poor reception. Moreover, the codec must also be able to deliver similarly high quality when processing other types of signal, such as music on hold. This challenge is anything



EVS developers Markus Multus, Guillaume Fuchs and Stefan Döhla (from the left) (Image courtesy: <https://www.fraunhofer.de>)

but simple, given that speech coding and audio coding are two separate worlds. The new codec therefore analyses the flow of signals every 20 milliseconds to distinguish between voice and music transmission, enabling appropriate algorithms to be applied.

Explains Dr Guillaume Fuchs, the research scientist who led the development of EVS at Fraunhofer, "The frequency range of the audio signals transmitted by currently available